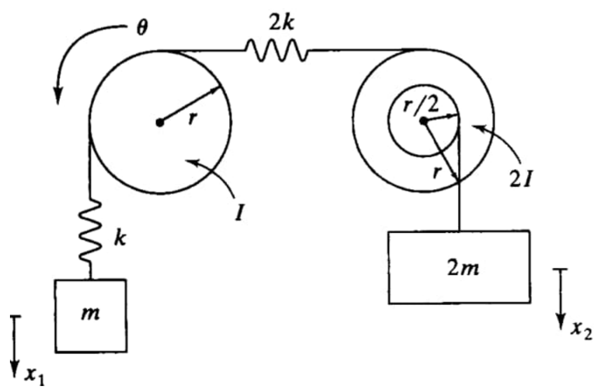
Focus of this homework:

- Concept of mass and stiffness matrices
- Application of Lagrange's equations to MDOF vibratory systems
- Vibration analysis of 2DOF and MDOF systems

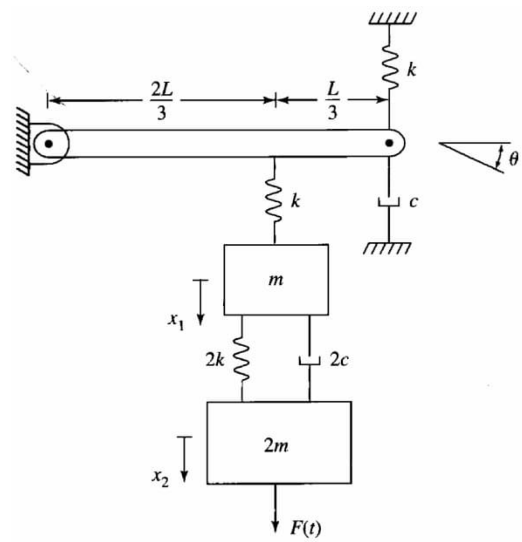
Submit homework (in hard copies) by the beginning of the class. It is your responsibility to make sure that homework is submitted in good quality. Please organize your work neatly. Suggest to start each problem on a new page.

1. For each of the two systems below, assume small oscillations and obtain the mass, stiffness, and damping matrices from energy method. For system (b), the mass of the rigid bar is also m. Use the Lagrange's equations to obtain the governing equations of motion for large amplitudes of oscillation, and then linearize to obtain the governing equations of motion around the equilibrium positions. Cast the linearized equations in matrix form.

(a)



(b)



2. Rao 5.31

5.31 Two identical pendulums, each with mass m and length l, are connected by a spring of stiffness k at a distance d from the fixed end, as shown in Fig. 5.35.

a. Derive the equations of motion of the two masses.
b. Find the natural frequencies and mode shapes of the system.
c. Find the free-vibration response of the system for the initial conditions $\theta_1(0) = a$, $\theta_2(0) = 0$, $\dot{\theta}_1(0) = 0$, and $\dot{\theta}_2(0) = 0$.
d. Determine the condition(s) under which the system exhibits a beating phenomenon.

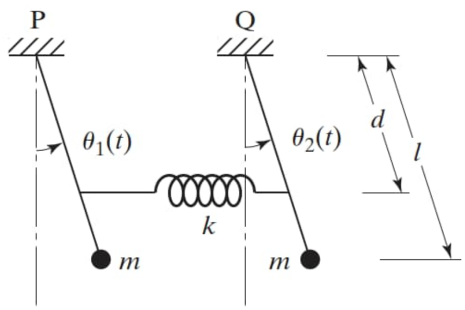


FIGURE 5.35 Two pendulums connected by a spring.

3. a) Derive the governing equations of motion of the two carts system. Use the parameters: $m = 500\ \mathrm{kg}, k = 1000\ \mathrm{N/m}, c = 100\ \mathrm{N\text{-}s/m}$.
   b) Is this a proportionally damped system? What are the modal damping ratios? Note: You have to first perform the eigenvalue problem analysis to obtain the natural frequencies and mode shapes.
   c) Find the free-vibration response of the system for the initial conditions: $y_1(0) = 2, \dot{y}_1(0) = 0, y_2(0) = 0, \dot{y}_2(0) = 0$.
   d) Obtain the free-vibration of the system numerically (using MATLAB or any suitable simulation software) and compare with the analytical result from part (c).

   [this problem is double the points]